Validation results - actual Logisim circuit

Five-Stage Pipelined Microcontroller for a Washing Machine
Logisim-Evolution 5.0.0 • Assignment 2

Simulator: supplied Logisim Evolution 5.0.0 all-JAR (unmodified), Java 21.0.2 (latest layout rerun). Tests use its CircuitState/Propagator and Pin APIs to stimulate the actual gates, wires, memories and registers. This is not a reference Processor model. Sim.java and validate.py reproduce the observations; every raw sample is retained in tests/*.csv.

All tests start with RESET=1 for one global rising edge, then RESET=0. Unlisted inputs begin at zero. Sensors are changed with CLK low and held through subsequent rising edges. The safety test also samples ACTUAL immediately after opening the door, before a clock. 17 automated checks passed.

1 Groups 1-3: hazards, control and memory

Test / initial conditions	Expected	Observed / result
8.1: R1=5,R2=3,R3=20; ADD R1,R2; SUB R3,R1	R1=8,R3=12; no stall	8 and 12; 0 stalls. PASS
8.2: RAM[10]=7,R3=5; LD R1,[R2]; ADD R3,R1	Exactly 1 stall; R3=12	1 stall; MEM/WB forwarding; R3=0C. PASS
8.3: R1=10,R2=3,R3=4,R4=12; ADD; SUB; ST	Store younger result 9 once	Only store (12,09); reload R5=09; 0 stalls. PASS
Source-use test: LD R1; LDI R1,9; immediate ST to F5	No false dependence on LDI destination; forwarded store address	0 stalls; DISPLAY=09. PASS
8.4: SUB to zero; BEQZ 07; wrong-path ST F4 and ADD	No actuator write or wrong-path ADD result	0 stores; target R3=55. PASS
Separate branch test: replace wrong-path ADD by HALT	Wrong HALT killed; target executes, later real HALT stops	R3=55; final HALTED=1; 0 stores. PASS
Store AB/load RAM[22]; ST/LD 80 and FF; ST/LD F1 with WATER=0	RAM=AB; reserved/unimplemented=0; RO sensor unchanged	R3=AB,R4=00,R5=00,R6=00; ACTUATORS=00. PASS

The raw traces explicitly include valid bits, stage PCs, stall/flush, both forwarding selectors, WB destination/data, MEM address/data and store enable. Architectural writes are sampled before their committing edge.

2 Groups 4-5: timer and safety

Test / stimulus	Expected	Observed / result
LIMIT=3; CTRL=3, then tick	Clear edge COUNT=0; hit at third increment	Cycle3 COUNT=0; cycle6 COUNT=3,MATCH=1. PASS
Disable after hit, then LIMIT=5; CTRL=3; tick260	Sticky match; second interval clear; modulo wrap	Before rearm COUNT=5,MATCH=1; cycle10=0/0; final COUNT=4,MATCH=1. PASS
CPU enables timer, then executes dependent LD/ADD and HALT	Timer advances on stall edge and while halted	Count increases by 1 across stall; halted count continues increasing. PASS
Accept START while closed/request0; request motor on	Actual motor on	ACTUAL=1, TRIP=0. PASS
Open door without clock, then clock once	Immediate actual off; edge latches trip	Before edge ACTUAL=0; after edge TRIP=1. PASS
Reclose door while request stays1	No automatic motor restart	TRIP=1, ACTUAL=0. PASS
Clear request; ACK with START still high (no new edge)	Rearm rejected	ACK_OK=0; TRIP=1. PASS
START low for edge; START high with closed door, request0; ACK	Explicit safe rearm clears trip	TRIP=0,ACK_OK=1; new motor request gives ACTUAL=1. PASS

An additional integrated wash run opened the door at cycle73: ACTUAL immediately changed1 to0, TRIP was1 after the next edge, and software reached DISPLAY=EE with requests=00. At cycle139 reclosure alone still left TRIP=1. START was then held low for two cycles and raised; software accepted rearm, and by cycle221 TRIP=0 and the new wash had ACTUAL=1. PASS.

Safety log entries can share a cycle number: these are deliberate combinational observations between clock edges. The door-open observation therefore proves immediate gating rather than relying only on the next rising edge.

3 Additional gate checks

ByteAdder: FF+01 -> SUM=00,COUT=1; 00+FF+carry1 -> 00,carry1. CMPGEU: 99>=100 ->0; 102>=100 ->1; 10>=9 ->1; 10>=11 ->0. SUB 00-01 ->FF. All passed.

4 Group 6: complete washing sequences

For each mode: reset; set DOOR_CLOSED=1, START=0, WATER=target-1; run 2 cycles; START=1; run110; change MODE to the other setting; run10; increase WATER above target; run260; set WATER=9; run300. TEMP/RPM remain0. Thus water inputs are stable over each MEM sampling edge, and MODE is deliberately changed after acceptance.

Observation	**MODE 00**	**MODE 01**
Initial / overshoot samples	99 -> 102	149 -> 153
Accepted MODE snapshot	0 (retained after pin becomes1)	1 (retained after pin becomes0)
Fill target / timer limits	100 / 80 / 40	150 / 120 / 60
First ACCEPT signal sample	Cycle14	Cycle14
First inlet-on sample	Cycle33	Cycle32
First pump-on sample	Cycle233	Cycle270
Drain completion input	9 (below threshold10)	9 (below threshold10)
HALTED first observed	Cycle456	Cycle480
Final actuator byte / display	40 / AA	40 / AA
Final physical motor / trip	0 / 0	0 / 0
Result	PASS	PASS

Observed actuator sequence in both runs includes 21 (fill), 20 (lock only), alternating 24/2C (wash directions), 30 (drain), 20 (pump off), 24 (spin), 20 (motor off), 40 (buzzer and unlocked). Both timer intervals are written and cleared independently. The stored limits match the originally accepted mode despite later pin changes.

5 Operator-panel verification

Main, the default opening circuit, was separately driven through its built-in global Clock using Logisim clock toggles. It completed MODE00 with DISPLAY=AA, ACTUATORS=40 and HALTED=1. The normal GUI run therefore uses the same wired controller as the direct-input tests.

6 Reproduction and scope

Open washing_pipeline.circ in Logisim Evolution 5.0.0; ROM is embedded. Follow README.txt for manual sensor stimulation or run the supplied Java/Python harness with your JAR path. This validation covers all six required groups. It does not claim exhaustive formal verification or a physical plant model. The following pages include supplied user screenshots, with their observations and scope stated explicitly.

7 Supplied screenshots and verification scope

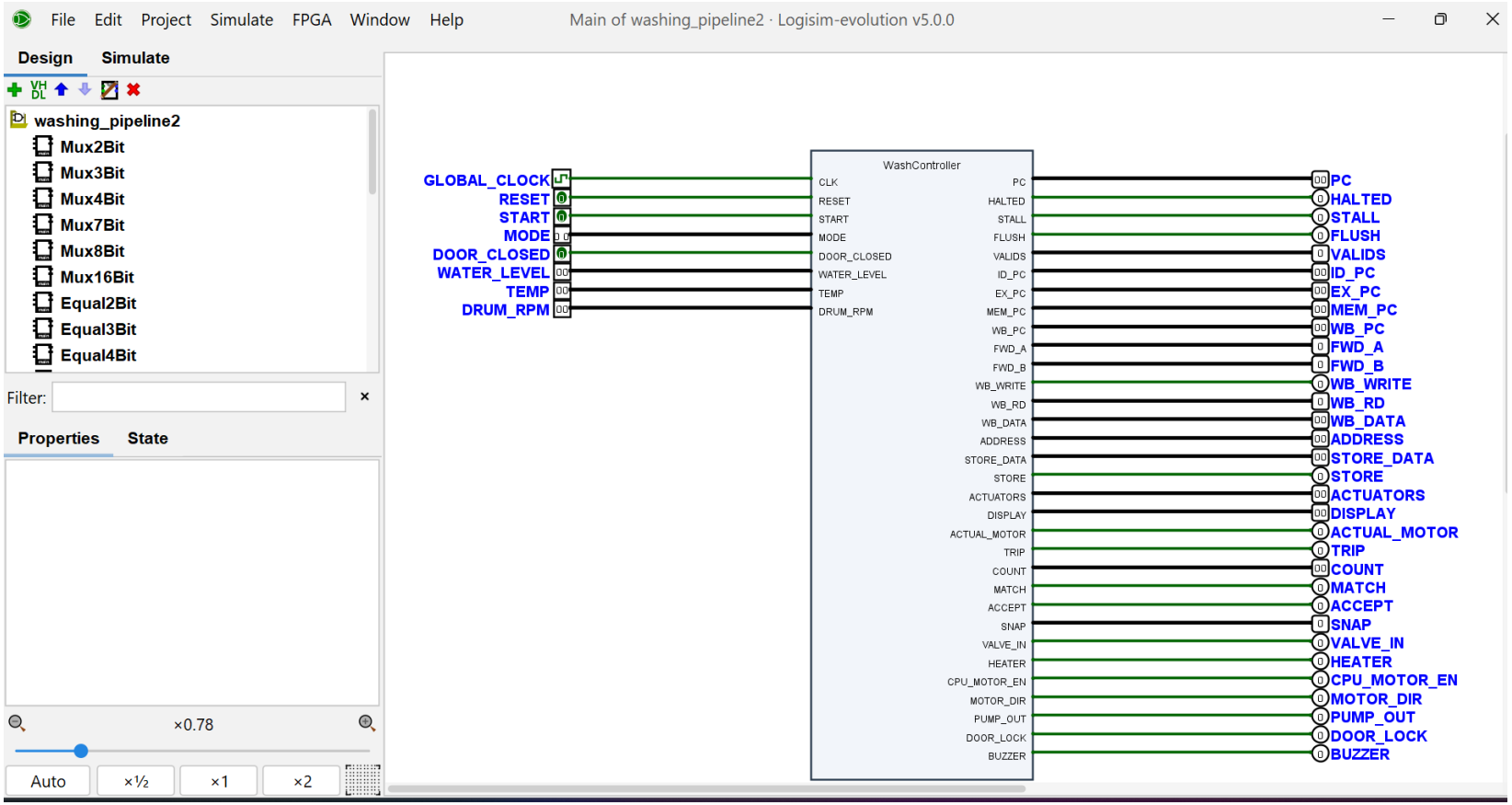
The following six captures were supplied by the user. Their project titles include washing_pipeline2 or washing_pipeline3 and some show an unsaved GUI state. They document those captured runs; they do not establish byte-for-byte identity with the final file. The latest layout-only revision was separately rerun using the actual Logisim engine: 17 functional groups and 1,096 register checks passed; 847 nets matched their expected connections; 285 component labels had no detected wire overlaps. The earlier user-device environment was Java/Javac 26.0.2 and Python 3.10.11.

Evidence	**Result**
Functional validation	17 groups passed
Register-bank checks	512 stored reads + 512 bypass + 64 write-disabled + 8 reset = 1,096 passed
Earlier user-device wiring audit	701 nets across 10 circuits; shown in terminal screenshots
Latest layout reference audit	847 nets across 22 internal circuits; includes mux/equality circuits
Manual captures	Initial panel, immediate door gate, sticky-trip abort, completion with retained mode

Screenshots are reproduced as supplied, including existing annotations. No signal values or test results have been altered. Each picture is captioned only with what it visibly supports. Earlier reference tables record the scripted stimulus sequence, not an inferred timing sequence from the still images.

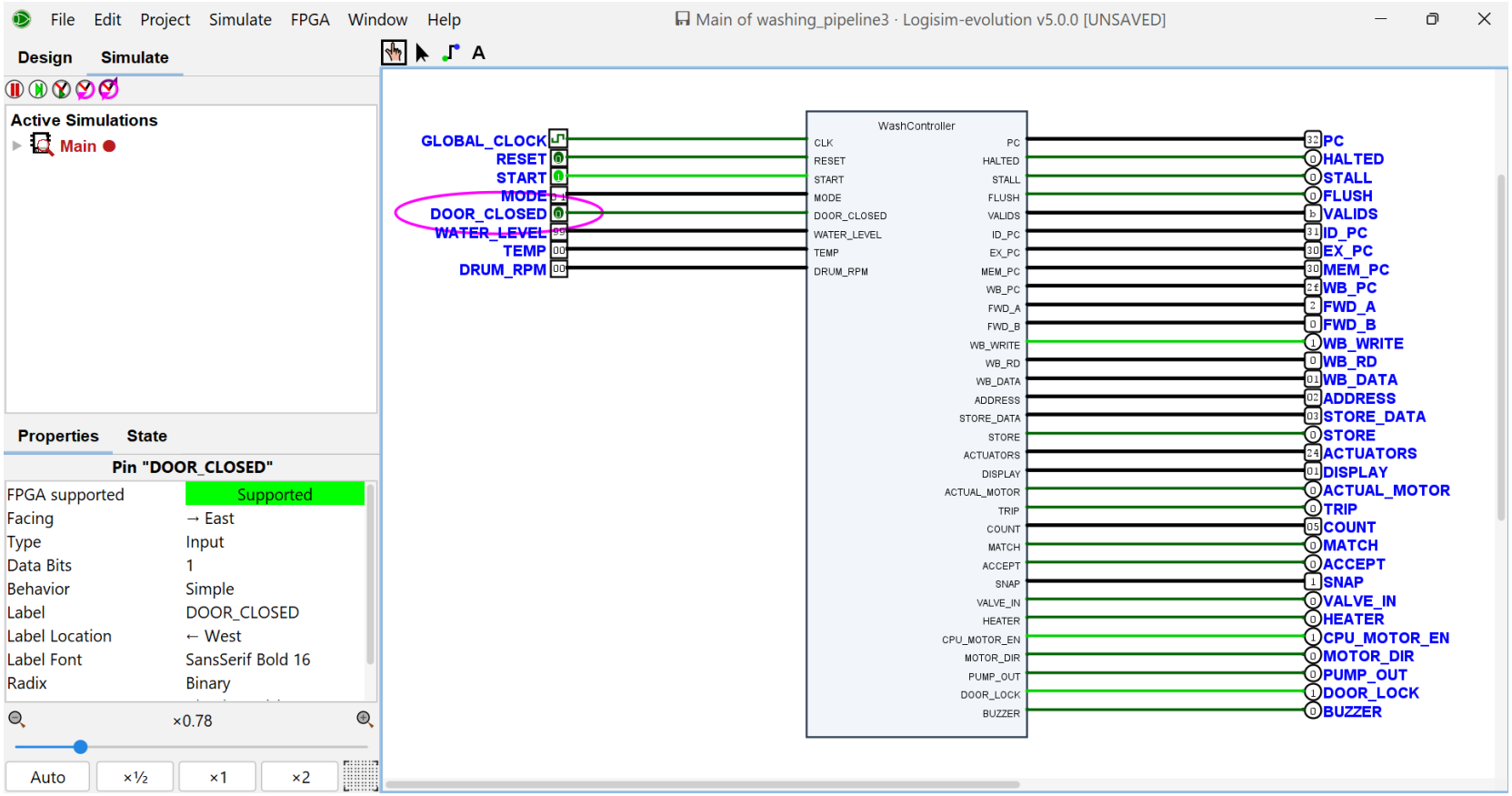
Initial operator panel

The supplied initial-state screenshot shows PC=00, START=0, DOOR_CLOSED=0 and inactive outputs. It documents the operator interface; it is not evidence of cycle completion.



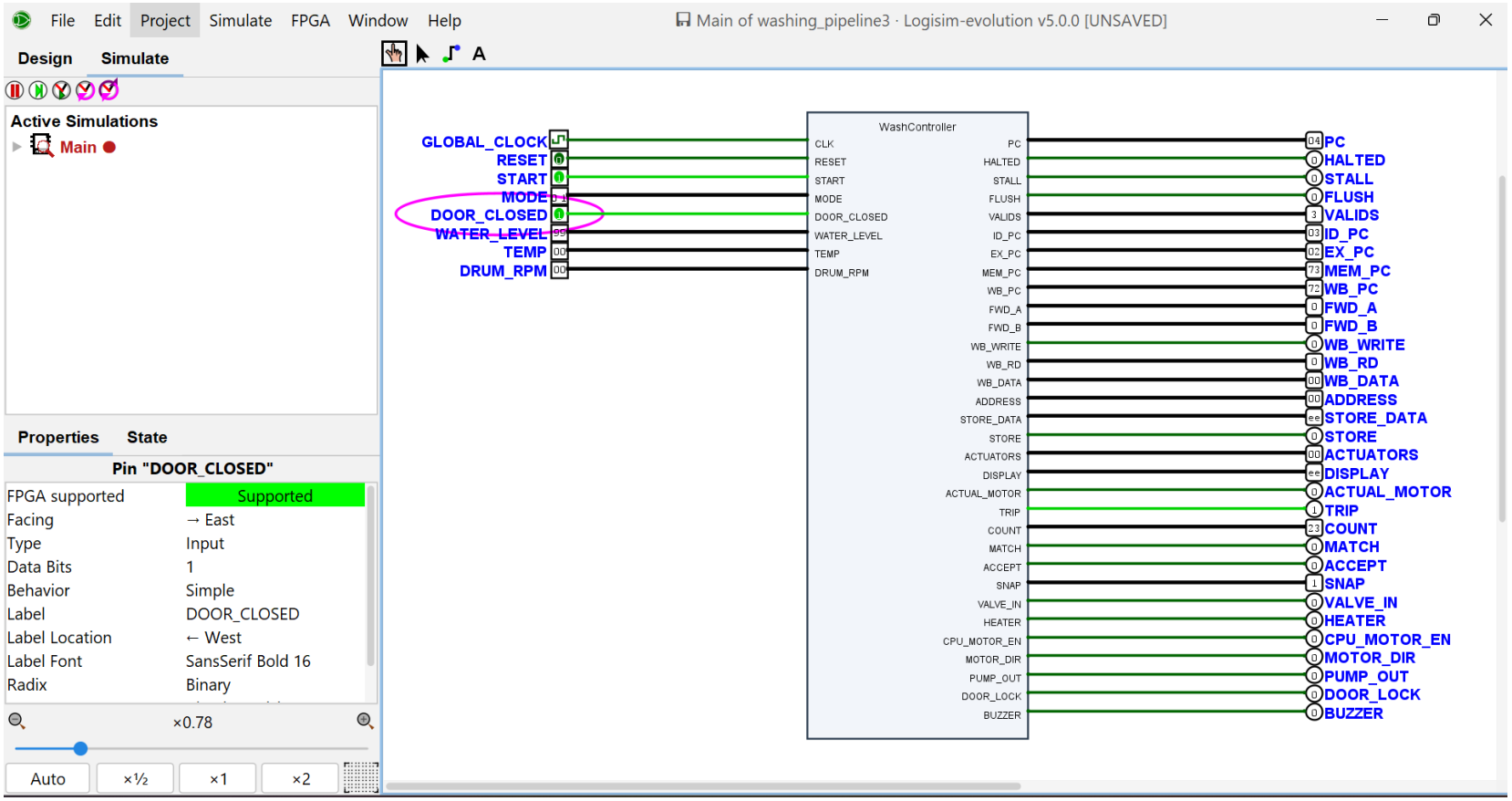
Immediate motor cutoff

DOOR_CLOSED=0 and CPU_MOTOR_EN=1, while ACTUAL_MOTOR=0. TRIP is still 0 in this frame, consistent with combinational door gating before the trip-latching edge. ACTUATORS=24 and DISPLAY=01 show the interrupted motor phase.



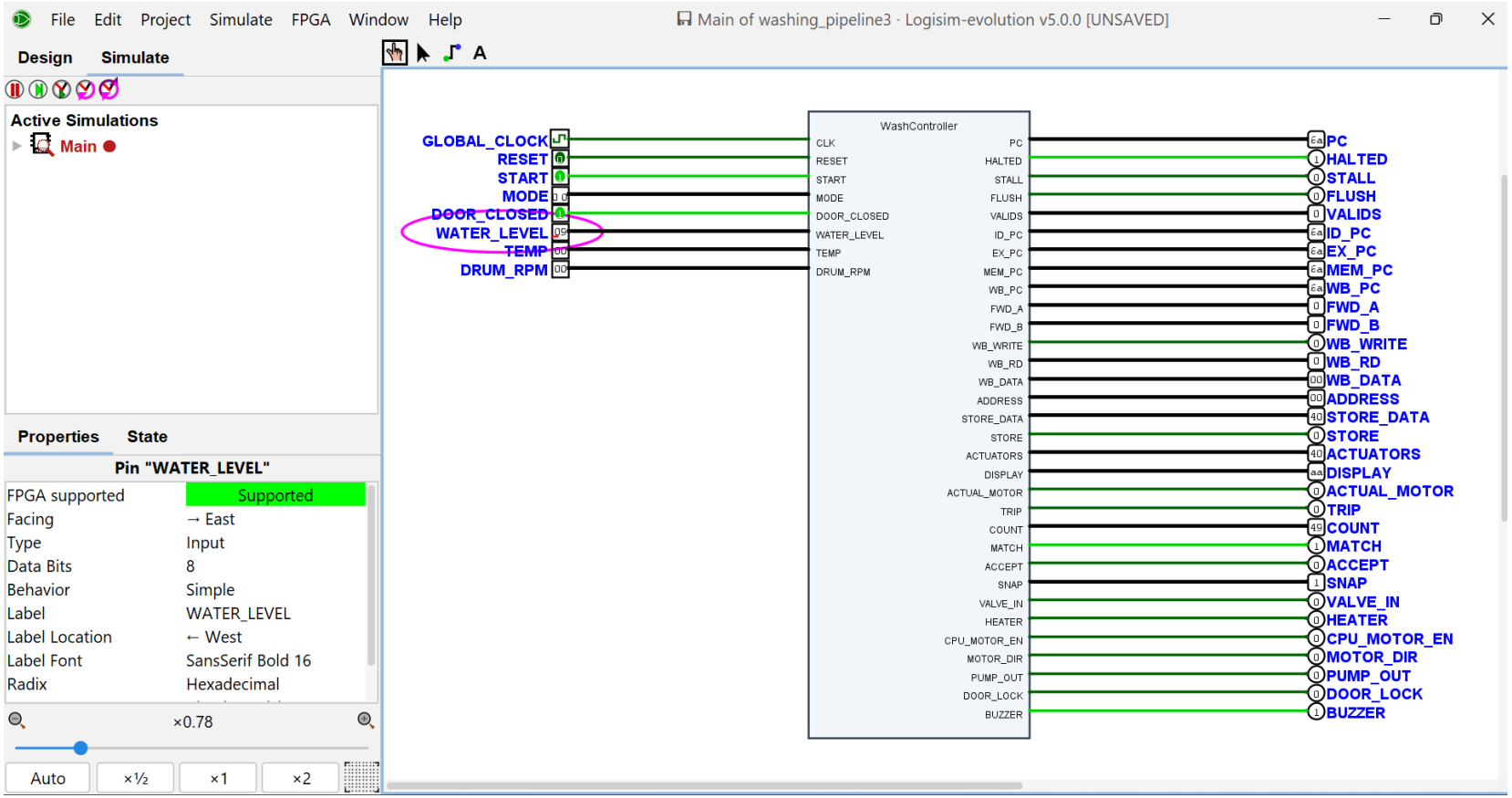
Reclosed door with a latched trip

DOOR_CLOSED=1, TRIP=1, ACTUAL_MOTOR=0, CPU_MOTOR_EN=0, ACTUATORS=00 and DISPLAY=EE. This shows that reclosure alone did not restart the motor and that software reached the abort state.



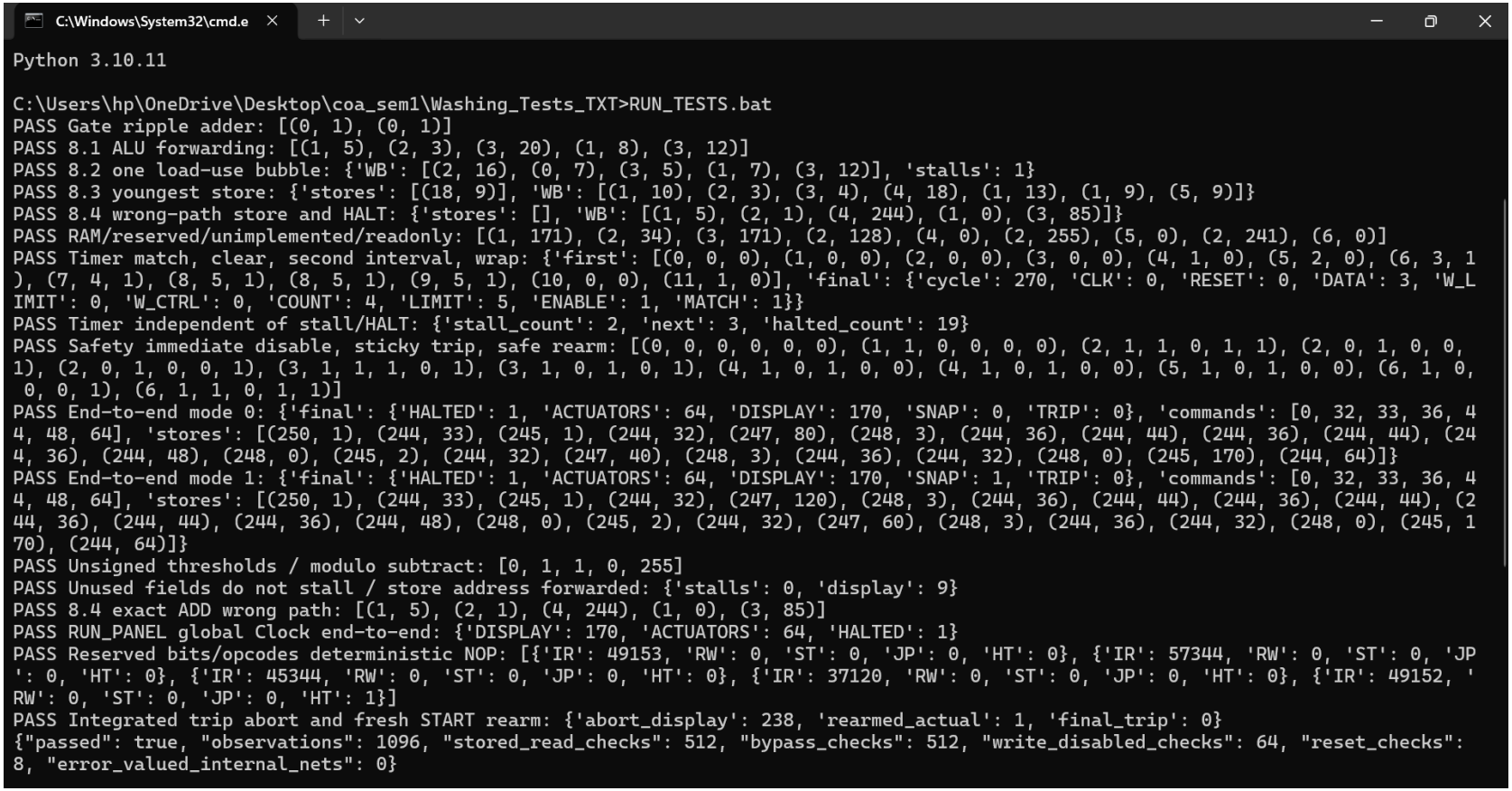
Completion and retained accepted mode

MODE input=00 while SNAP=1. HALTED=1, ACTUATORS=40, DISPLAY=AA, WATER_LEVEL=09, BUZZER=1 and ACTUAL_MOTOR=0. This frame supports completion with the earlier accepted mode retained despite a later input change.



User-device functional and register tests

The supplied terminal capture shows all 17 functional groups reporting PASS and 1,096 successful register-bank observations, including zero error-valued internal nets. Values in the terminal are decimal unless explicitly formatted otherwise.



User-device connectivity audit and completion

This capture shows the earlier audit of 701 nets across 10 circuits and the final successful launcher message. It predates the last layout-only revision. The revised-file reference audit checks 847 nets across all 22 internal circuits; this screenshot is not presented as a user-device rerun of that revision.

